IESE CONSULTING CLUB

CASE BOOK 2020

IN PARTNERSHIP WITH BOSTON CONSULTING GROUP

IESE Business School University of Navarra

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FORWARD

Dear Reader,

In partnership with Boston Consulting Group (BCG), we are proud to share the first edition of The IESE Case Book.

The IESE Consulting Club collaborated with BCG to create a competition for consulting club members at IESE (Class of 2020 and 2021) to write and submit original cases, to help the next generation of aspiring consultants prepare for interviews. Each submitted case passed through a careful and completely blind evaluation process conducted by the IESE Consulting Club and BCG, which selected the 7 best cases based on pre-defined evaluation criteria. These 7 best cases form the IESE Case Book which will enable the reader to:

- Develop interview guidance techniques to practice peer-to-peer preparation
- Improve problem-solving skills
- Mimic real-life interview situations with solid cases, accompanied with BCG insights

We have paid special attention to the user experience so that the whole process of practicing peer-to-peer mocks becomes easier, seamless, and intuitive. We have tried our best to provide a detailed explanation of how to use the book and some tips for interviewer guidance throughout all cases.

We would like to thank Boston Consulting Group for their partnership and support in bringing this case book to life. Wishing you the best of luck and hope you enjoy your preparation journey!

IESE Consulting Club

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CONTENT

Acknowledgements

The first edition of the IESE Case Book is the result of coordinated efforts between Boston Consulting Group and all IESE students that employed time, creativity, and dedication in bringing this case book to life.

The collection of cases that you will find in the following pages were created as part of the 1st IESE Casebook Writing Competition amongst consulting club members, who were challenged to write cases based on business problems that could help consulting candidates in their preparation process.

Each one of the cases submitted passed through a careful and completely blind evaluation process, conducted by the IESE Consulting Club and BCG, which selected the seven best cases based on pre-defined evaluation criteria.

We would like to thank the BCG Team for their time, support, and guidance in this endeavor, the participants for the time and effort they put to develop their cases, and the IESE Casebook Team members from Class of 2020 and 2021 who helped bring this book to life in its final format.

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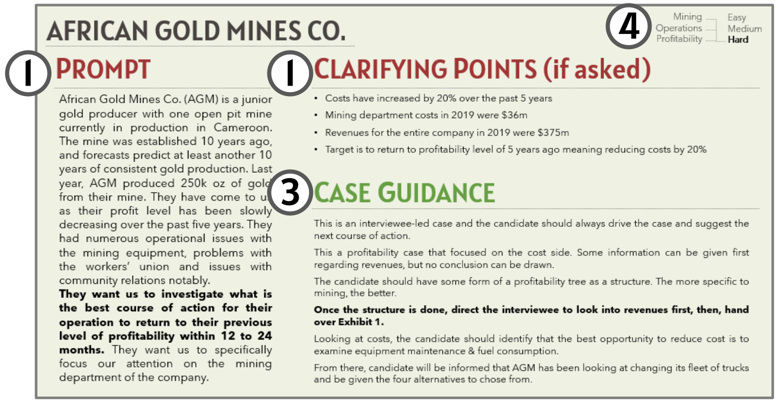
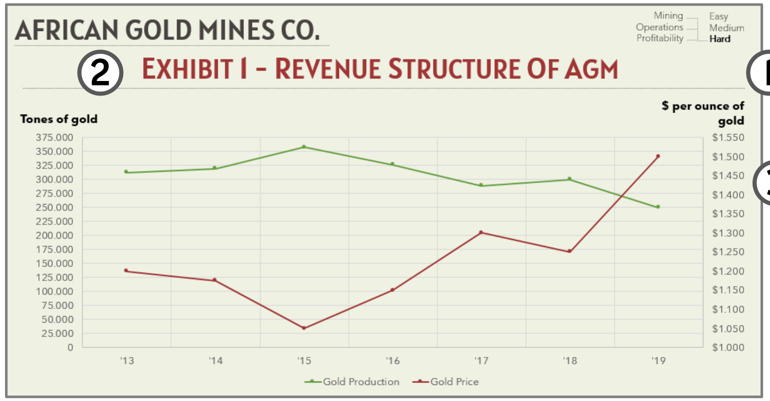
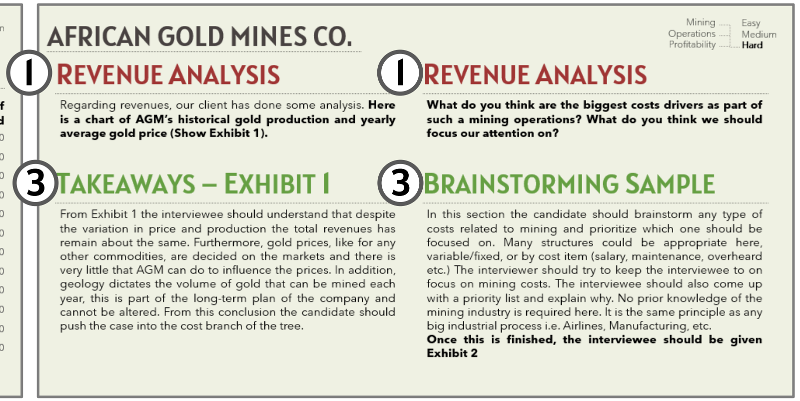
How To Use This Case Book







We designed this book to be practical and straightforward for both the interviewer and the interviewee. Read the following instructions to ensure a smooth application process and to extract most value out of this casebook.

1. Red titles mean information that the interviewer has to give to the interviewee, including the prompt, clarifying questions and exhibits.
2. Exhibit pages provide necessary information to interviewees solve the cases and should be handed in their entirety when instructions asked to do so
3. Green titles mean information that can help the interviewer in guiding the case including expected takeaways, expected considerations, calculations and sample recommendations. Interviewers should not disclose this information to candidates but use it to guide themselves into the flow of the case and help candidates in navigating the numbers.
4. Each case is classified by its industry, theme, and concept tested as well as by its level of difficulty.

COWBON EMISSIONS

By Emily Hilton (IESE MBA 2021)

Agriculture
Sustainability
Operations

Easy
Medium
Hard

Agriculture
Sustainability
Operations

Easy
Medium
Hard

COWBON EMISSIONS

PROMPT

Our client is a major milk producer in New Cowland, Milking it Co., MIC. New Cowland has recently introduced a law that means MIC has to reduce its GHG emissions by 45% of 2019 levels within the next five years or face being shut down or heavily fined. They currently produce 20% of New Cowland's GHG emissions.

The CSO has hired us to figure out a way to reach this target.

CLARIFYING POINTS (if asked)

- MIC produces approximately 100% of New Cowland's milk supply
- Does not have plans for expansion but reducing volumes is not an option
- Their only product is milk and have no plan to diversify
- Budget for this project is $750m/year for the next five years (for perspective, current revenues are $15billion)
- MIC owns the entire production chain – from farms, production and transport, they sell to a variety of clients

The candidate might ask what the breakdown of where GHG comes from within the business, this is shown in Exhibit 1

- 5% of their market is local, the rest is foreign
- Only need to reduce GHG directly produced by MIC

CASE GUIDANCE

This case is designed to test brainstorming, business decision making skills and logic. It will help candidates wanting to practice market sizing and working on unconventional problems. For calculations ignore the time value of money.

It is a long case designed for advanced candidates; some aspects can be removed for the sake of time – these are clearly marked.

Interviewer guidance has been provided at various stages.

COWBON EMISSIONS

INTERVIEWER GUIDANCE – STRUCTURE

A good framework would touch on:

- Different ideas to reduce GHG emissions in the different segments of the business (farming, processing, transportation, overheads, etc.)
- Ability to implement changes
- Financial implications
- Non-financial implications – PR, risks (e.g. change in government, backlash from farmers)

The candidate should lead the case towards understanding the current breakdown of GHG emissions – if not the interviewer should gently nudge toward this path.

GHG EMISSIONS ANALYSIS

The candidate should be presented with Exhibit 1 if they have asked about the breakdown of GHG in each business segment. They should be told that MIC has already taken measures to reduce Overheads Emissions and MIC believes that there is no further that can be done to reduce these.

EXHIBIT 1 TAKEAWAYS

- GHG emissions have been increasing over the last three years with farming being the key driver
- Farming is the largest emitter followed by production
- From farming, cows are by far the largest emitter producing 63% of 2019's emissions – this would be the key thing to look into first as it can make the biggest difference (it is the only value that is over 50% of the emissions by itself)
- Processing is the second biggest emitter so that should be focused on next (20% of total emissions)

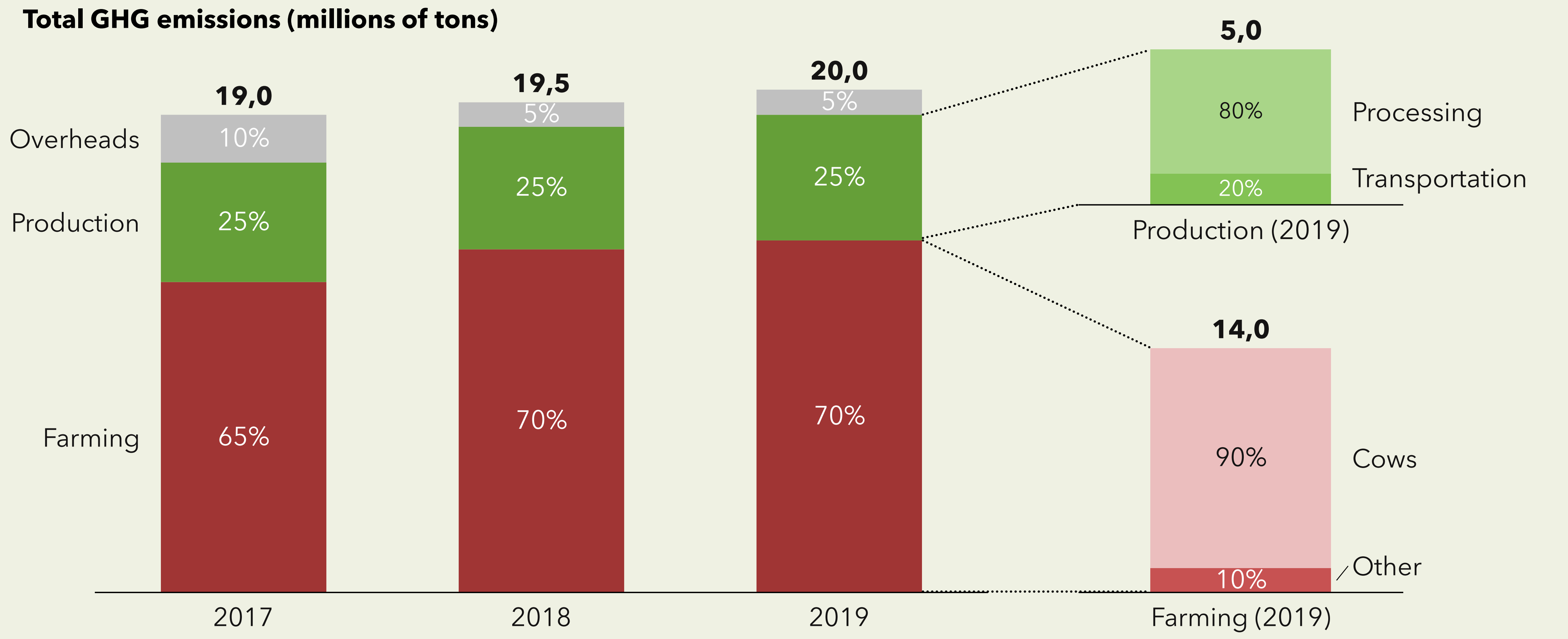
The candidate should identify cows as the first logical step to explore in reducing emissions.

COWBON EMISSIONS

Agriculture
Sustainability
Operations

Easy
Medium
Hard

EXHIBIT 1 – TRENDS OF GHG EMISSIONS OF MIC



COWBON EMISSIONS

Agriculture
Sustainability
Operations

Easy
Medium
Hard

Cow Emission Analysis

There is a new breed of cow that MIC is interested in to replace their current herd. All characteristics are the same as the current herd, but they produce 33% less GHG emissions.

As MIC owns farms all over New Cowland, they want us to approximate how many cows they have in total.

Estimation Data

Provide only if requested:

- Population of New Cowland: 5 million
- Milk produced per cow: 10L/days
- Number of days in a year: 300
- Domestic market: 5% of sales
- Domestic market share: 100%
- Average milk consumption per person: 150L/year

Suggested Calculation

**** if the interview is progressing slowly, skip this estimation and give the number of cows of 5 million, then move straight to part 2 after reading the prompt below****

This is an estimation problem – the candidate should recognize that they should size the herd using suitable assumptions, provide the estimation data only if the candidate asks.

Total annual domestic consumption:
[Milk consumption per person]*[population of NZ]
= 750 million L/year
Cows required for domestic consumption:
[Annual dom. Cons.]/([Daily pro./cow]*[days in a year])
= 250,000 cows
Total cows:
[domestics cows]/[% of market]
= 5 million cows

COWBON EMISSIONS

COW EMISSION ANALYSIS

Knowing that there are 5 million cows in the MIC herd and the average life of a dairy cow is 5 years. **Calculate the total cost of replacing the herd and the total emissions saved from this replacement.**

- The old breed cost $2000 and the new breed costs $2500 to purchase
- The old breed emits 2.5 ton/year and the new breed emits 33% less GHG

SUGGESTED CALCULATION & TAKEAWAYS

	YEAR 1	**YEAR 2**	**YEAR 3**	**YEAR 4**	**YEAR 5**
Cost	[1 million cows] *[$500/cow] = $500 million	[1 million cows] *[$500/cow] = $500 million	[1 million cows] *[$500/cow] = $500 million	[1 million cows] *[$500/cow] = $500 million	[1 million cows] *[$500/cow] = $500 million
GHG ton reduction	[1 million cows]* [0.33*2.5t] Or [12.6million tons]* [33%]/[5/1] = 0.825 mil tons	[2 million cows]* [0.33*2.5t] Or [12.6million tons]* [33%]/[5/2] = 1.65 mil tons	[3 million cows]* [0.33*2.5t] Or [12.6million tons]* [33%]/[5/3] = 2.475 mil tons	[4 million cows]* [0.33*2.5t] Or [12.6million tons]* [33%]/[5/4] = 3.3 million tons	[5 million cows]* [0.33*2.5t] Or [12.6million tons]* [33%]/[5/5] = 4.125 mil tons
% GHG reduction	4.125%	8.25%	12.375%	16.5%	20.625% (~21%)

Candidate can calculate only Year 5 as this is the final number required but must check that annual budget is not exceeded.

Candidate should conclude that this is approximately halfway to our goal, spending 67% of our budget. Next should look at reducing emissions from production.

COWBON EMISSIONS

PRODUCTION EMSSIONS ANALYSIS

What are some ways that the GHG emissions can be reduced during the production phase?

BRAINSTROMING SAMPLE

**** if the interview is progressing slowly, skip this brainstorming and move straight to part 2 after reading prompt 2 on this page****

This brainstorm is a chance for the candidate to be creative. No structure is superior but looking at options split between processes and transportation is one option. Keep pushing until you are satisfied with the ideas generated.

PROCESSES	TRANSPORTATION
Alternative power sources (all renewable)	Larger trucks (i.e. less GHG/L transported)
Carbon capture of emissions	Electric vehicles
More energy efficient processes	Shorter routes of transportation
R&D into different ways to process milk	Regular maintenance to increase efficiency
Outsource production	Outsource transportation
Shutdown inefficient plants	

PRODUCTION EMSSIONS ANALYSIS

MIC has been looking into different ways to reduce GHG emissions in both processes and transportation and they have identified these options (show Exhibit 2).

CALCULATION AND TAKEAWAYS – EXHIBIT 2

This exhibit is intentionally challenging to understand, point the candidate towards the footnotes if they are struggling to understand the meaning of each column.

Key insights from Exhibit 2:

- Conversion to renewable energy has the largest emissions savings (potential reduction of 14% of GHG) and can be completely replaced within the 5-year period
- Replacing sterilizing units will only reduce emissions by 4% total and will take 20 years to carry out
- The other processes are also not cost effective and should be written off straight away
- Transportation is a cost-effective method of GHG reduction which can be completed within 5 years and will reduce emissions by 5% total

	COST PER YEAR	TOTAL COST (5 YEARS)	GHG REDUCTION PER YEAR	TOTAL GHG REDUCTION (5 YEARS)
Renewable energy source	$150 million	[$150m] * [5years] =$750 million	[25%]*[80%]*[70%] *[20%]= 2.8% (0.56 mil tons)	[2.8%] * [5 years] = 14% (2.8 mil tons)
New Sterilizing Process	$100 million	[$100m] * [5years] =$500 million	[25%]*[80%]*[20%] *[5%]= 0.2% (0.04 mil tons)	[0.2%] * [5 years] = 0.8% (0.2 mil tons)
Transport electrification	$50 million	[$50m] * [5years] =$250 million	[25%]*[20%]*[100%]*[20%]= 1% (0.2 mil tons)	[2.8%] * [5 years] = 5% (1 mil tons)

The candidate should conclude that conversion to renewable energy and electrifying the transport fleet will be the most efficient costing $200m/year and reducing the GHG emissions by 19% in total by year 5

Exhibit 2 – Alternatives To Reduce GHG Emissions Of MIC

	PROPOSED CHANGE TO PRODUCTION	TOTAL % REDUCTION OF EMISSIONS[1]	# OF UNITS[2]	MAX. % ANNUAL UNIT REPLACEMENT [3]	ANNUAL TOTAL COST OF MAX REPLACEMENT[4]
Process	Renewable energy source	70%	10	20%	$150m
	New Sterilizing Process	20%	20	5%	$100m
	New Packaging process + material	2%	50	20%	$50m
	Other	0.1%	500	10%	$1m
Transport	Electrification of fleet	100%	500	20%	$50m

1 % of reduction in GHG emissions that the process or transportation will emit, based on the current emissions shown in Exhibit 1, once 100% of the units have been replaced

2 number of units MIC currently owns that can be replaced in a 1:1 ratio with the new alternatives

3 % of units that can be replaced annually (due to end of life requirements) e.g. a total of 2 units can be replaced per year for 'renewable energy source'

4 How much it will cost MIC annually to replace the maximum allowable units e.g. will cost MIC $150m to replace 2 units of 'renewable energy source' per year

COWBON EMISSIONS

RECOMMENDATION

Great, the CSO is about to join us, can you please provide her with a brief summary of what we have discussed today?

SAMPLE RECOMMENDATION

The candidate should summarize that:

- We should replace the herd with new breed to reduce emissions by 21% in year 5, costing $2.5 billion
- We should convert to renewable energy and replace our transport fleet reducing our emissions by 19% in year 5, costing $1 billion
- These actions will get us 90% towards our target of 45% reduction
- We have $50m/year of our budget left to figure out the last 10% which could include.... (any ideas that you have discussed e.g. carbon credits, hiring a lawyer to reduce potential fines, other emission reductions)

A great candidate would also briefly discuss any risks or next steps:

- Adaption to new technology
- Chance new breed has challenges
- Views from employees and farmers about changes
- Chance of lobbying government so law is reversed
- Next steps: contact breeders to make sure they have enough cows, look for other reduction methods, etc.

AFRICAN GOLD MINES CO.

By Marc-Oliver Granger (IESE MBA 2021)

Mining
Operations
Profitability

Easy
Medium
Hard

AFRICAN GOLD MINES CO.

Mining · Operations · Profitability

Easy · Medium · Hard

PROMPT

African Gold Mines Co. (AGM) is a junior gold producer with one open pit mine currently in production in Cameroon. The mine was established 10 years ago, and forecasts predict at least another 10 years of consistent gold production. Last year, AGM produced 250k oz of gold from their mine. They have come to us as their profit level has been slowly decreasing over the past five years. They had numerous operational issues with the mining equipment, problems with the workers' union and issues with community relations notably.

They want us to investigate what is the best course of action for their operation to return to their previous level of profitability within 12 to 24 months. They want us to specifically focus our attention on the mining department of the company.

CLARIFYING POINTS (if asked)

- Costs have increased by 20% over the past 5 years
- Mining department costs in 2019 were $36m
- Revenues for the entire company in 2019 were $375m
- Target is to return to profitability level of 5 years ago meaning reducing costs by 20%

CASE GUIDANCE

This is an interviewee-led case and the candidate should always drive the case and suggest the next course of action.

This a profitability case that focused on the cost side. Some information can be given first regarding revenues, but no conclusion can be drawn.

The candidate should have some form of a profitability tree as a structure. The more specific to mining, the better.

Once the structure is done, direct the interviewee to look into revenues first, then, hand over Exhibit 1.

Looking at costs, the candidate should identify that the best opportunity to reduce cost is to examine equipment maintenance & fuel consumption.

From there, candidate will be informed that AGM has been looking at changing its fleet of trucks and be given the four alternatives to chose from.

AFRICAN GOLD MINES CO.

Mining Operations Profitability — Easy Medium Hard

REVENUE ANALYSIS

Regarding revenues, our client has done some analysis. **Here is a chart of AGM's historical gold production and yearly average gold price (Show Exhibit 1).**

TAKEAWAYS – EXHIBIT 1

From Exhibit 1 the interviewee should understand that despite the variation in price and production the total revenues has remain about the same. Furthermore, gold prices, like for any other commodities, are decided on the markets and there is very little that AGM can do to influence the prices. In addition, geology dictates the volume of gold that can be mined each year, this is part of the long-term plan of the company and cannot be altered. From this conclusion the candidate should push the case into the cost branch of the tree.

REVENUE ANALYSIS

What do you think are the biggest costs drivers as part of such a mining operations? What do you think we should focus our attention on?

BRAINSTORMING SAMPLE

In this section the candidate should brainstorm any type of costs related to mining and prioritize which one should be focused on. Many structures could be appropriate here, variable/fixed, or by cost item (salary, maintenance, overheard etc.) The interviewer should try to keep the interviewee to on focus on mining costs. The interviewee should also come up with a priority list and explain why. No prior knowledge of the mining industry is required here. It is the same principle as any big industrial process i.e. Airlines, Manufacturing, etc.

Once this is finished, the interviewee should be given Exhibit 2

Exhibit 1 – Revenue Structure Of AGM

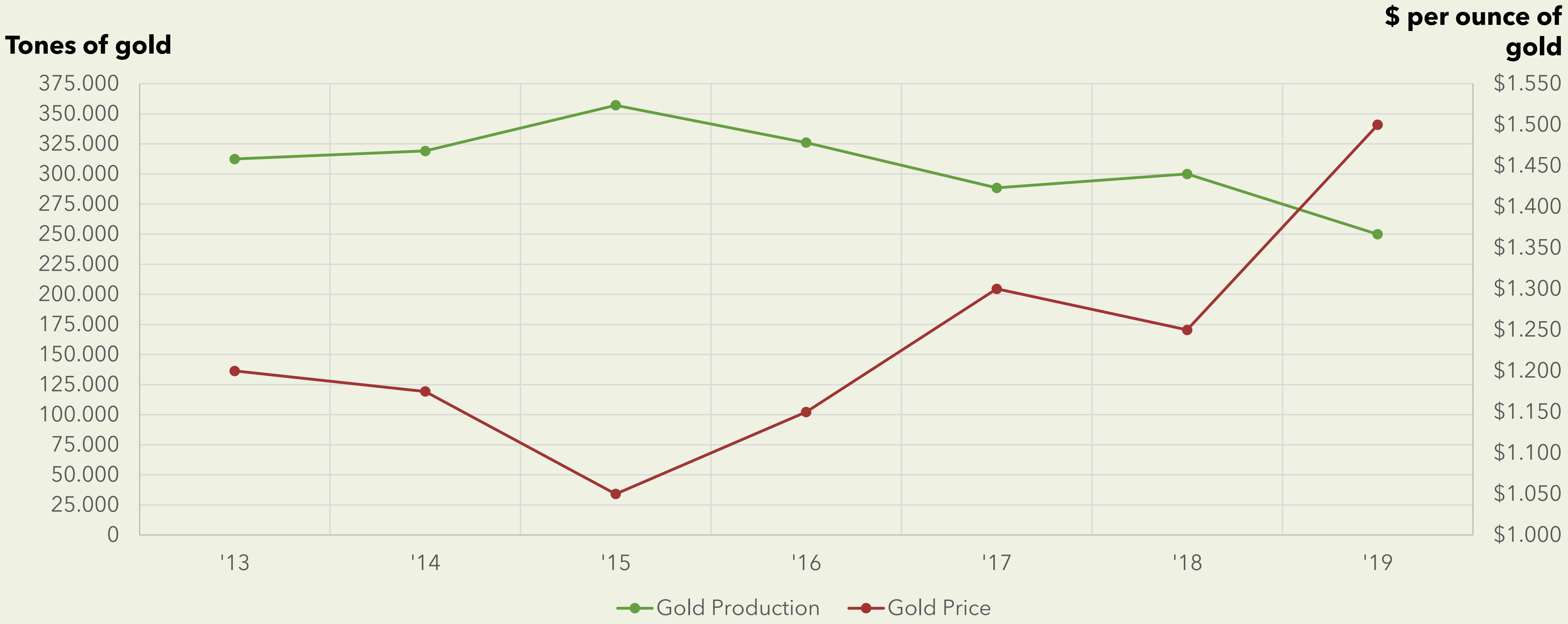
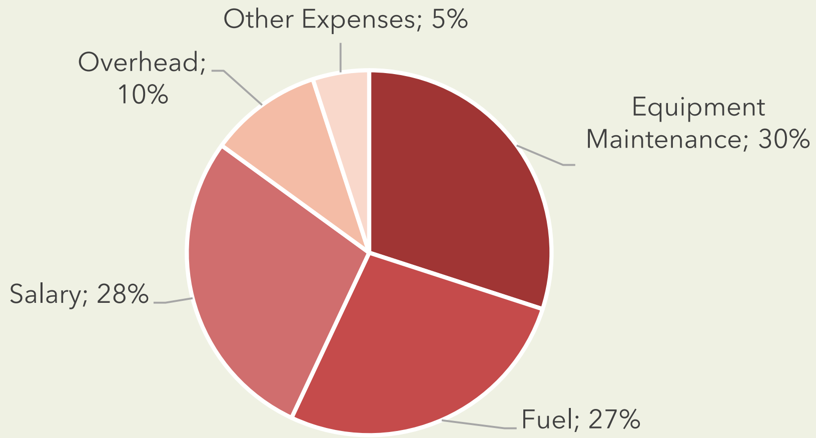
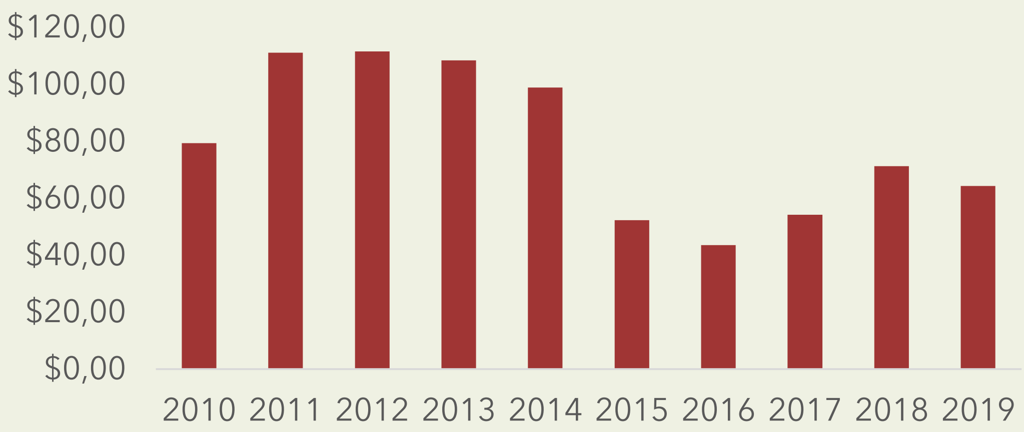


Exhibit 2 – Cost Structure Of AGM Mining Department





COSTS VARIATION YOY AT AGM MINING OPERATIONS

Costs	'15	'16	'17	'18	'19
Salary	1%	0%	-2%	-1%	1%
Maintenance	3%	9%	9%	12%	19%
Fuel	3%	3%	5%	9%	13%
Overhead	1%	1%	2%	0%	1%
Other	1%	5%	5%	0%	0%

AFRICAN GOLD MINES CO.

TAKEAWAYS – EXHIBIT 2

Exhibit 2 is purposely very busy to evaluate how the candidate can associate information from different sources together.
In the end the candidate should quickly identify maintenance costs and fuel costs as the two main drivers of the drop in profitability of AGM, but also acknowledge that salary is a big cost. **Once the candidate has indicated s/he would like to investigate maintenance/fuel, give Exhibit 3.**

TRUCK SELECTION ALTERNATIVES ANALYSIS

Our client has been looking at replacing its fleet of mining trucks as they are getting very close to the end of their useful life. They done some analysis and found four possible alternatives. Show exhibit 3.

TAKEAWAYS – EXHIBIT 3

The interviewee should compare the financials of the four alternatives. The exhibit also give the state of the current situation for comparison purposes. The four alternatives are:

1. Overhaul of the current fleet by keeping the same trucks but changing major components.
2. Buying a new fleet of the exact same truck.
3. Buying a fleet of a new model of diesel trucks from a different supplier.
4. Buying a fleet of new semi-electric autonomous trucks from the same supplier that require less fuel and no driver to operate.

AFRICAN GOLD MINES CO.

CALCULATION – EXHIBIT 3

ALTERNATIVES	FUEL COST PER YEAR	MAINTENANCE PER YEAR	CAPEX PER YEAR	TOTAL COST PER YEAR
Current Situation Caterpillar 777E	20*300*95 = $570k	$540k	$100k	570+540+100=$1210k
Overhaul of Current Caterpillar 777E	20*300*80 = $480k	$500k	600/5= $120k	480+500+120=$1100k
New Caterpillar 777E	20*300*60 = $360k	$340k	1000/10= $100k	360+340+100=$800k
Komatsu HD785-7	20*300*70 = $420k	$450k	1200/12= $100k	420+450+100=$970k
Caterpillar 780D Autonomous Semi-Electric	20*300*40 = $240k	$360k	1600/8= $200k	240+360+200=$800k

FURTHER INFORMATION (if requested)

- Trucks are working 20 hours per day
- Trucks work 300 days per year
- Price of Fuel is 1$/L
- The mine uses 20 trucks
- Specification and capacity of all trucks are similar. They all can do the job
- Ignore the time value of money
- Assume straight line depreciation
- Candidate should also compare the new alternative to the current to see if it fulfills our client requirement of reducing cost by 20%. Total costs = 36m -> 20%*36m = 7.2m
  Old situation – New = $1210k-$800k = $410k per truck x 20 trucks =Savings of 8.2m per year. Regardless of which option the candidate chooses, it meets our client requirement

EXHIBIT 3 – TRUCK SELECTION ALTERNATIVES FOR AGM

ALTERNATIVES	CAPEX	MAINTENANCE COSTS PER YEAR	FUEL CONSUMPTION	EXPECTED LIFE
Current Situation: Caterpillar 777E	$0.1m	$540k	95 L/hr	1 yr
Overhaul of Current Caterpillar 777E	$0.6m	$500k	80 L/hr	5 yrs
New Caterpillar 777E	$1.0m	$340k	60 L/hr	10 yrs
Komatsu HD785-7	$1.2m	$450k	70 L/hr	12 yrs
Caterpillar 780D Autonomous Semi-Electric	$1.6m	$360k	40 L/hr	8 yrs

Truck Selection Criteria/Risks – Brainstorming

From the calculation of the four alternatives the interviewee should understand that both the new 777E trucks or 780D autonomous truck are financially the same. **The interviewer should prompt the interviewee to brainstorm on criteria/risks that our client should consider when making decision.** This brainstorming could be structured in many forms, but Pros & Cons would be appropriate.

Sample Brainstorming

	PROS	**CONS**
777E	• -Same model the client as been using • Same spare parts / supply chain • Same maintenance processes • Good training of employees • Longer life 10 years	• Possible higher wage costs • Possible higher operation costs if fuel prices increase • Potentially more dangerous options • Operational efficiency might be less • Require continuous training of operators
780D Autonomous	• Lower fuel consumption, costs saving could be more important if fuel prices go up • Further cost saving possible if employees can be laid off • Increased safety benefits • Potential increased operational benefits since these trucks could be more efficient • No training required for operators • Lower fuel consumption = Lower GHG emissions, more environmentally friendly option	• New technology which might be harder to implement in a very remote site in Africa. • Technical support might be harder to obtain • Maintenance and operation personnel will require training, and this might be hard to obtain in a remote setting • New spare parts / supply chain required • Old inventory might not be usable • Laying-off employees in a small village in Africa might not be appreciated by the communities • Shorter life 8 years

AFRICAN GOLD MINES CO.

RECOMMENDATION

Our client is coming to our office in a couple of minutes and we need you to make them a recommendation.

SAMPLE RECOMMENDATION

In this case there is no one good answer. Both the 777E and the 780D options are financially equal and both could be defended. The interviewee should use the result of the financial analysis and the brainstorming to make a sound and logical argumentation for which option is chosen.

The recommendation should include:

- The interviewee should structure the recommendation by starting that the AGM has a cost problem mainly based on the high maintenance cost and fuel consumption of the old trucks.
- To reduce their costs back to their previous level the company should change its fleet of mining trucks.
- Indicate which option she/he chooses and back that up with logical arguments from the brainstorming.
- Acknowledge some of the risks related with that option.

NICA PRODUCTIONS

By Alfonso Tomás Durandeu (IESE MBA 2021)

Media & Entertainment
Profitability
Operations Cost

Easy
Medium
Hard

NICA PRODUCTIONS

Media & Entertainment
Profitability
Operations Cost

Easy
Medium
Hard

PROMPT

Nica Productions is an American Media company that is trying to figure out which will be its next project. This company has an extensive experience producing series and movies for all type of audiences and has got many awards doing so. This company has two alternatives, produce a series for a streaming company or a movie to be projected in cinemas worldwide.

Produce media content implies big investments and low certainty about the potential incomes, which depends on many factor, for that reason, our client has hired us to help him/her decide which is the best alternative for him/her.

CLARIFYING POINTS (if asked)

- There is no specific profitability goal
- The company is worldwide known with access to top star directors, actors and technical staff
- It has not budget limitation
- Both alternatives look for a worldwide reach but target different type of customers
- Production of any alternative will last one year
- There is no alternative project
- The main source of revenue of both projects depends on audience

CASE GUIDANCE

This is a quantitative case that requires the candidate to estimate the potential cashflow of different alternatives in order to get the NPV and decide the best option for the client

The candidate will need to ask for additional information that is necessary to solve the problem, rather than relying on the interviewer to dispense it.

Especially for less finance-minded interviews, you may have to help nudge trough the math and formulas

NICA PRODUCTIONS

INTERVIEWER GUIDANCE – STRUCTURE

The candidate should express that the company will **pursuit the project that generate the higher positive profits** and show that in his/her Framework.

- A good candidate will take in consideration the uncertainties related with production and revenues streams and express the intention to estimate the NPV of each project.
- Other aspects to bear in mind are: Competition, Market trends, Company's strategy, etc.

If asked for information about **REVENUES and COST, make him/her BRAINSTORM about it**

- A good candidate would understand that revenues not only come from tickets or broadcasting royalties but also from merchandising, games, DVD/Blu-rays, etc. Respect the cost, the candidate should mention the basic: Director, Actors, production, marketing, etc.

NICA PRODUCTIONS

Media & Entertainment
Profitability
Operations Cost

Easy
Medium
Hard

PROJECT ALTERNATIVES ANALYSIS – MOVIE

The company is not sure about **which type of movie they want to produce and if it will be able to hire the director and actors for the desired alternative**. However it was able to assign probabilities for each scenario. Income will be based on the audience which depends on the critics received. **Which project (movie or series) should our client choose? (Show Exhibit 1).**

ANALYSIS TAKEWAYS - MOVIE

In order to estimate the NPV of this alternative the candidate should **calculate the corresponding Cash Flows and request for a Discount Rate (20%).** The investment are made at the beginning of the project and incomes are received at the end of year 1 (Solution in the following slide).

A good candidate will:

- read the note to get the information about how much money receive the company from the Cinema chains
- be structured and present the calculation with clarity (Desire: decision tree)

NICA PRODUCTIONS

ANALYSIS CALCULATION – MOVIE

TOTAL AUDIENCE (TA):
TA = [(AudienceGood * %Good) + (AudienceBad * %Bad)]
TA NOLAN = [(600M * 70%) + (400M * 30%)] = [420M + 120M] = 540M
TA BAY = [(400M * 80%) + (200M * 20%)] = [320M + 40M] = 360M

INCOME:

1ST YEAR
There is no income

2ND YEAR
Income = Total Audience * Ticket Price * %Commission
Income NOLAN = 540M * 10 USD/t * 10% = 540M USD
Income BAY = 360M * 10 USD/t * 10% = 360M USD

INVESTMENT:
Each alternative has it own cost structure (See exhibit 2)

	ALTENATIVE 1		ALTERNATIVE 2	
	GOOD	BAD	GOOD	BAD
Probability (%)	70%	30%	80%	20%
Audience (M)	600	400	400	200
Contribution (M)	420	120	320	40
Average audience	540		360	

(M USD)	ALTERNATIVE 1		ALTERNATIVE 2	
Year	0	1	0	1
Investment	-150	0	-100	0
Income	0	540	0	360
Cash Flow	-150	540	-100	360
NPV	-150	450	-100	300
Accumulated NPV	**300**		**200**	

	ALTENATIVE 1	ALTERNATIVE 2
Probability (%)	60%	40%
NPV (M USD)	300	200
Contribution (M)	180	80
Average NPV (M USD)	**260**	

Exhibit I: Movies Alternatives

	Alternative 1		Alternative 2	
Target Audience	Teenagers and Young Adults		Kids and Teenagers	
Probability (%)	**60%**		**40%**	
Director	Christopher Nolan	25M USD	Michael Bay	10M USD
Main Actor	Tom Hardy	50M USD	Mark Walhberg	25M USD
Support Actor	Michael Cain	25M USD	Tyrese Gibson	5M USD
Others	Production & Marketing	50M USD	Production & Marketing	60M USD
	Critics		**Critics**	
Good	70% chance of getting 600M audience		80% chance of getting 400M audience	
Bad	30% chance of getting 400M audience		20% chance of getting 200M audience	

Note: Filmmakers receive as income 10% of the ticket sales (10 USD/ticket);

NICA PRODUCTIONS

PROJECT ALTERNATIVES ANALYSIS - SERIES

The company already have the **script for a potential Series**, and in case of choosing this alternative it will receive an upfront from the Streaming company. Income will be based on the Audience per episode in the first year **(Show Exhibit 2).**

ANALYSIS TAKEWAYS - SERIES

In order to estimate the NPV of this alternative the candidate **calculate the corresponding Cash Flows and request for a Discount Rate (20%).** The investment are made at the beginning of the project and incomes are received at the end of year 1 (Solution in the following slide)

A good candidate will:

- understand that the Scrip expense (10M USD) is a sunk cost and it should not be taken in consideration to make the decision
- be structured and present the calculation with clarity (Desire: decision tree)

If the candidate realize about the sunk cost, she/he will choose to produce a Series. If not, make him/her identify the mistake

	NPV	**NPV (with sunk cost**
Movies	30%	40%
Series	100	70

NICA PRODUCTIONS

Media & Entertainment
Profitability
Operations Cost

Easy
Medium
Hard

ANALYSIS CALCULATION – SERIES

TOTAL AUDIENCE (TA):

TA = [(AudienceGood * %Good) + (AudienceMedium * %Medium) + (AudienceBad * %Bad)] * #Episodes

TA = [(100M * 30%) + (70M * 40%) + (40M * 30%)] * 10 = [30M + 28M + 12M] * 10 = 700 M

INCOME:

1ST YEAR

The income in the first year is the Upfront = 20M USD

2ND YEAR

Income = Total Audience * Income per episode (PMV) = 700M * 600k USD = 420M USD

INVESTMENT:

If the company chose this alternative, it should make an additional Investment of 100M USD (The Script is a sunk cost)

ADDITIONAL DATA	
Upfront (M USD)	20
# episodes	10
Income per episode (PMV) (USD)	600,000

INVESTMENT (USD)	
Cost (M USD)	100
Upfront (M USD)	20
Investment (M USD)	80

	GOOD	MEDIUM	BAD
Probability (%)	30%	40%	30%
Audience (M)	100	70	40
Contribution (M)	30	28	12

(M USD)	SERIES	
Year	0	1
Investment	--100	0
Income	20	420
Cash Flow	--80	420
NPV	--80	350
Accumulated NPV	**270**	

EXHIBIT 2: SERIES

EXPENSES

Script (already bought)	10
Actors	60
Production	40

ADDITIONAL DATA

Upfront (M USD)	20
# episodes	10
Income per episode (PMV)	600,000

AUDIENCE LEVEL	AUDIENCE PER EPISODE (M USD)	PROBABILITY
High	100	30%
Medium	70	40%
Bad	40	30%

Project Alternatives Analysis – Additional Info

Would your decision change with this new information? (Show Exhibit 3)

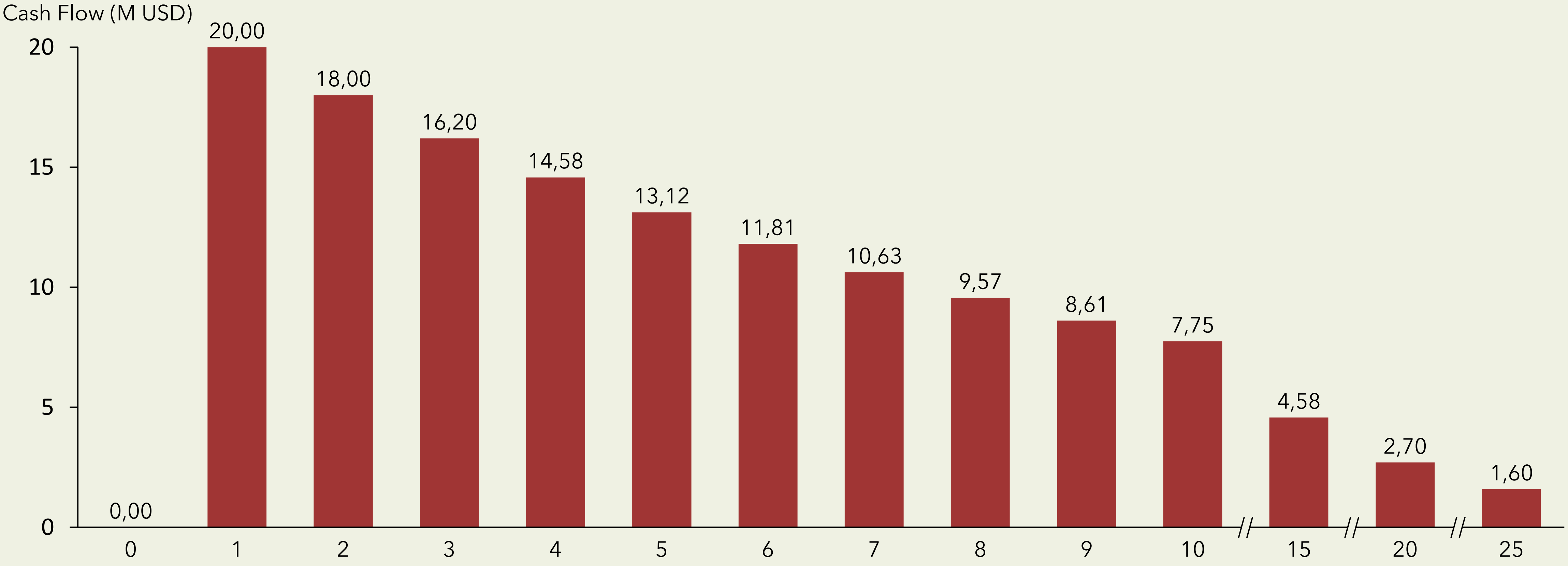
Takeways – Exhibit 3

Exhibit 3, which contains a cash flow for Michael Bay movie from merchandising and other incomes.
The candidate should **identify the tendency in the cashflow (CAGR -10%) and calculate a perpetuity.**
Adding those Cash Flows to Alternative 2 will affect the NPV.
As result of this information the final decision will change.

	PERPETUITY
Initial Cash Flow (M USD)	20
Growth (%)	-10%
Discount Rate (%)	20%
NPV (M US)	**66.7**

	ALTERNATIVE 1	**ALTERNATIVE 2**
Probability (%)	60%	40%
NPV (M USD)	300	266.6
Contribution (M)	180	107
Average NPV (M US)	**287**	

EXHIBIT 3 – MOVIE ALTERNATIVE 2



PROJECT ALTERNATIVES ANALYSIS – OTHER FACTORS

Which other factor would you take into consideration?

EXPECTED CONSIDERATION

The candidate should quickly identify the uncertainty and our capability to assess it as a main risk.
Factors that can influence:

- Do not finish the production process on time and miss the target launch
- The launch of a good movie made by a competitor
- An economic crisis that affects the consumption
- Others

This question is to assess candidate business sense and creativity. There is room for the interviewee to discuss other factors, including other revenues streams and intangible factors. Always justifying his/her answer.

Recommendation

Great, our client is coming and will request a recommendation?

Sample Recommendation

The candidate should be concise and structured, without mentioning topics that were not discussed.

It is important to highlight the uncertainty of the decision process and he/she should suggest potential way to reduce it.

PIPELINE OIL TECHNOLOGY

By Roberto Carlos De Araujo (IESE MBA 2021)

Oil & Gas
Operations
Pricing

Easy
Medium
Hard

PIPELINE OIL TECHNOLOGY

Oil & Gas
Operations
Pricing

Easy
Medium
Hard

PROMPT

Minerva's University Fluids Research Lab has discovered a more efficient way to transport crude petroleum oil inside pipelines. This new technology can be used in midstream applications where the oil is acquired from the extraction plant and delivered to the refinery plant. The university invested $ 1,200M in this project during the last 12 years.

The new technology mixes water and oil under certain conditions to reduce the loss of energy, caused by the friction between the oil and the pipeline surface while being transported. As a result, the transport between two given points gets 15% faster and the useful lifetime of the pipelines increases by 20%.

Minerva's University asked our help to value for how much they should sell the technology.

CLARIFYING POINTS (if asked)

What is the market?
- Mexico.
How big is the market?
- 4,800 km of pipeline.
Who are the competitors and market share?
- National Oil Company (NOC) is the only player. **However, the market is open for the last two years.**
Which are the potential buyers?
- Primarily, NOC. However, other two prospects are interested in entering the market.
Does the university have a patent? How long does it last?
- The university has already filed the patent, which lasts for 20 years.

What is NOC pipelines current capacity?
- Full capacity. Surplus is transported by more expensive means such as rail car, barge and truck.
How is the demand for crude oil?
- NOC sells all the crude oil it buys. See Exhibit 2 for the next years forecast.
Can NOC build more pipelines to substitute other means of transportation?
- Yes, it is an alternative. However, there are costs involved. See "4. Given Data - Alternative: Expand Pipeline Network"
How long does it take to implement this technology?
- Minerva's University estimates that the technology would be running in 100% of the pipelines in one year at $2,000 M installation cost.

CASE GUIDANCE

Crude oil Value chain
Downstream:
- Exploration - Extraction
Midstream:
- Transportation
Upstream:
- Refinery - Marketing/Selling

Midstream Business Model

1) Buy from the extractors (*)
2) Transport crude oil from the extractor to the refinery
3) Sell to the refineries at $10 per barrel

(*) Assume that Cost of transportation already incorporates the buying price from the extractors.

PIPELINE OIL TECHNOLOGY

INTERVIEWER GUIDANCE – STRUCTURE

Identify the main drivers of the pricing and evaluate the results by comparing to other investments.

A) PRICING:
Value = (1) Savings Costs of transportation + (2) Savings Cost of replacing pipelines – (3) Cost to implement

(1) Savings in Cost of transportation
- Increase in flow speed leads to increase in the pipeline's delivery capacity
- Save costs of sales by switching from other means of transportation to pipelines

(2) Savings in Cost of replacing pipelines
- Save Costs by using the pipelines for longer, reducing the replacement

(3) Cost of installation
- $2,000 M

B) COMPARISON: Compare the value of the technology to other investments to validate the final pricing

(4) Research Investment (by Minerva's University):
- $1,200 M (spent in the last 12 years by Minerva's University)

(5) Alternative investment (for National Oil Company):
- Investment to expand the pipeline network from 2 M to 2.5 M barrels per day

PIPELINE OIL TECHNOLOGY

Oil & Gas
Operations
Pricing

Easy
Medium
Hard

REVENUE ANALYSIS

Handle over Exhibit 1 if the interviewee asks for revenues and/or costs breakdown

DEMAND ANALYSIS

Handle over Exhibit 2 if the interviewee asks about the demand for the next years

PRICING CALCULATION

(1) Savings in cost of transportations (SCOT):

SCOT = (I) Sales for 20 years * (II) Savings in switching transportation = $162,000 M * 0.035 = $5,670 M ~ $6,000 M in 20 years (or $300 M / year)

(I) Sales for 20 years = Chart area (trapezoid) * 360 days * price per barrel: = [(2+2.5) * (20) x (1/2)] x 360 x 10 = $162,000 M

(II) Savings in switching transportation: = % increase pipeline use * proportional savings = 10% * 0.35 = 0.035

Switching from truck (5%) and rail (5%) to pipeline: % increase pipeline use = 15% * 70% = 10.5% ~ 10%

Proportional Savings: = [% truck * (cost truck – cost pipeline) + % rail * (cost rail – cost pipeline)] = 5% * ($6 - $2) + 5% * ($5 - $2) = 0.35

TIP: ROUND UP / DOWN

Recommend to the interviewee to round up some numbers to facilitate the calculations. Some suggestions are underline in the case. Nonetheless, strong candidates figure out these opportunities by themselves.

EXHIBIT 1

Income Statement

+ Sales	$7,200 M
- Cost of transportation (*)	$1,980 M
- Cost of replacing pipelines	$1,000 M
- Other operating costs (**)	$3,500 M
Operating profit	$720 M

Table 1: Income statement for Year 0, assuming sales of 2M barrels per day.

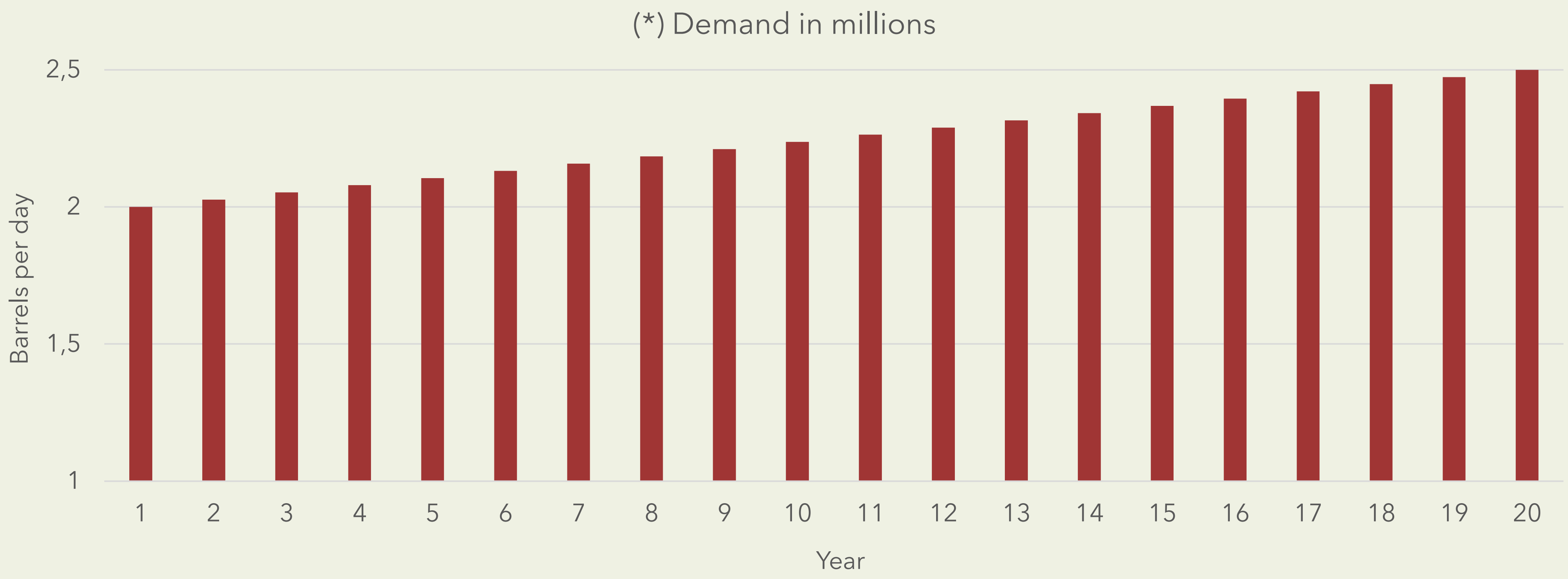
Means	Cost	Barrels
Pipeline	$2 per barrel	70%
Truck	$6 per barrel	5%
Barge	$3 per barrel	10%
Rail	$5 per barrel	15%

Table 2: Cost of Sales breakdown per means of transportation

(*) Cost of transportation already includes the buying price from the extractors.
(**) Assume that Other Operating Costs does not change over the years.

EXHIBIT 2

Crude oil demand forecast



(*) In a given year, assume that the demand per day is equal for all the 360 days.

PIPELINE OIL TECHNOLOGY

PRICING CALCULATION

(2) Savings in Cost of replacing pipelines (SCRP):

= Cost of replacing pipelines per year * 20 years * (1 - new lifetime/current lifetime)
= $1,000M * 20 * [1 - 4/5 years] = $4,000M in 20 years (or $200M per year)

GIVEN DATA:

- Cost of replacing pipelines: $1,000M per year
- Lifetime of regular pipeline: 5 years

ALTERNATIVE ANALYSIS

Expand network pipeline

Investment to expand the pipeline network:

= (Additional capacity in 20 years / Additional Capacity) * cost to implement
= (2.5 M – 2M) / 0.05 M * $700 M = $7,000 M

GIVEN DATA:

- Additional capacity: 50,000 barrels per day
- Time to implement: 2 years
- Cost per additional capacity = $ 700 M

PIPELINE OIL TECHNOLOGY

Oil & Gas
Operations
Pricing

Easy
Medium
Hard

COMPARISON ANALYSIS

PRICING	**$8,000 M**
(+) Savings in Cost of transportation	$6,000 M
(+) Savings in Cost of replacing pipelines	$4,000 M
(-) Cost of installation	$2,000 M

INVESTMENTS	
(-) Research Investment (University)	$1,200 M
(-) Alternative investment (National Oil Company)	$7,000 M

NEGOTIATION

The University can recover its investment selling the technology for any price over $1,200 M. Given the estimated savings and cost of installation, NOC will pay less than $ 8,000 M to guarantee profits/savings.

Alternatively, NOC can construct its own pipeline network for a total investment of $7,000 M. Then, it is better to buy the technology for $ 7,000 or less than to go forth with this alternative investment. Note that the alternative investment is limited by additional 50,000 barrels/day in two years, while the new technology can be put in operation in just one year.

Therefore, a reasonable price for selling this technology would be between $1,200 M and $7,000 M.

PIPELINE OIL TECHNOLOGY

RECOMMENDATION

What would be your final recommendation to Minerva's University?

SAMPLE RECOMMENDATION

The general recommendation is open. One of the possibilities is to sell the new technology for a price of $1,500 M plus a 30% participation in the additional revenue while the patent holds (40% of $10,000M = $3,000M in 20 years, disregarding cost of installation). There are three reasons that support this proposal:

1) The saving costs by increasing the volume of crude oil transported in pipelines are relevant. According to the calculations, $300 M per year (around 40% of the current Operating Profits)

2) The improvement in the pipeline lifetime is also relevant accounting for $200 M per year (around 25% to 30% of the current Operating Profits)

3) The alternative of expanding the pipeline network is a higher investment than the saving costs generated by the new technology. Besides that, expanding the pipelines is limited by additional 50,000 barrels/day in two years, while the new technology can be put in operation in just one year.

A great candidate would also briefly discuss any risks or next steps:

Main risks / sensitive assumptions:
- Delay in the installation of the new technology
- Limitation in reaching some regions, since it can be done only by a specific mean of transportation
- High investment to implement ($2,000M). Options: cash surplus, bank loan, increase in equity
- Crude oil price fluctuation -> Use future contracts to guarantee buying and selling prices
- Demand changes because of crisis or other external factor

Next steps:
- Confront its own calculation with NOC's calculations/data to validate the assumptions
- Define a negotiation strategy based on the calculations/assumptions
- If the negotiation fails, look for other prospect buyers

THE BOOKSTORE

By Víctor Manzanares Bonilla (IESE MBA 2021)

Retail
E-commerce
Market Entry

Easy
Medium
Hard

THE BOOKSTORE

Retail
E-commerce
Market Entry

Easy
Medium
Hard

PROMPT

Our client, Classic Bookstore (CB), is a traditional bookstore chain in Spain, specialized in non-technical books. CB revenues have stagnated for the past 3 years, with a stable and loyal customer base. Now, as part of their new growth strategy, they are considering whether to enter the electronic books market.

The client is considering selling their own reading devices and developing a website to sell the e-books for this device.

Our client asked us to analyze this opportunity and provide a recommendation.

CLARIFYING POINTS (if asked)

- CB only sells to customers through physical stores, no online business is available.
- CB has stores in the biggest cities of Spain.
- CB has 3 types of customers: AVID READERS (2 books/month), OCCASIONAL READERS (1 book/2 months) and RARE READERS (1 book/6 months).
- Non-technical physical book market in Spain has been stagnated for the past 3 years.
- CB has reached an agreement with an e-Reader manufacturer in China. Total cost per device would be 60€. These devices can only support the e-books sold in CB's new website.
- CB has no specific growth rate in mind and are opened to suggestions from us.

CASE GUIDANCE

This is a case designed to be led by the candidate. Start by reading the case question and let the candidate drive the analysis. Do not provide any information until is asked.

This case primarily tests the understanding on market entry and its implications on the current business model.

For simplicity, taxes and value of money over time have been ignored in this case, although excellent candidates should mention them during the case.

Interviewer Guidance – Structure

Candidate's structure should cover the following key aspects of the problem.

MARKET OPPORTUNITY: What is the market **size** of generic e-books in Spain? What is this market's **growth**?

POTENTIAL SHARE: What would be our **market share**? How many competitors are we facing in this market?

POTENTIAL PROFIT: What is the potential **profit** of this new market? Expected revenues vs expected costs? What **investment** is required to enter this new market? What is the expected **return of investment** of our client? Payback period?

CAPABILITIES & RISKS: Does this new market align with our client's **strategy** and **capabilities**? Do they have the know-how required? Have they got the financial capabilities to undertake this investment? What is the potential **cannibalization** of this new business model with the current one?

The analysis should be led by the candidate, starting for the market size. When the candidate asks about information of the market and size, ask him to estimate the size of the non-technical books in Spain, both in paper and e-books.

INTERVIEWER GUIDANCE – MARKET SIZING

(Suggested approach)

1. Population of Spain: 45M

2. Target population that reads: We assume people from 15 to 80 years old.
Population 0 – 20 (25%): 11.25M (Population 15-20: 11.25/4=2.8M)
Population 21 - 40 (25%): 11.25M
Population 41 – 60 (25%): 11.25M
Population 61 – 80 (25%): 11.25M
Total target population = 2.8 + 11.25 + 11.25 + 11.25 = 36.5 M

3. Percentage of population that buys books: We assume 60% of people between 15 and 80.
Total target population = 36.5M x 0.60 = 22M people

4. Type of readers: We assume the following categories and quantities
Avid Readers: 12 books/year x 2M readers = 24M books/year
Occasional Readers: 6 books/year x 5M readers = 30M books/year
Rare Readers: 2 books/year x 15M readers = 30M books/year
Total books: 24 + 30 + 30 = 84M books

5. Percentage of paper books and e-books and average prices:
PAPER BOOKS (93% of books): 78M books x 15€/book = 1,170M€
E-BOOKS (7% of books): 6M books x 8€/book = 48M€

TOTAL MARKET: 1,170 + 48 = 1,220M €

THE BOOKSTORE

PROFITABILITY ANALYSIS – GIVEN DATA

In the next step, the candidate should focus on profitability. **Hand Exhibit 1** for this part and **provide the following information if requested:**

REVENUES

- Market Annual Growth: 5%
- Expected Market Share: 1%
- Paper book Gross Margin: 33%
- E-book Gross Margin: 40%
- E-reader price: To be determined by candidate with **Exhibit 2 and 3.** Price range should go between 60 and 100€ to compete against Kindle.
- No customers change from paper to e-book after year 1. New readers are coming from new customers.

COSTS

- Webpage Investment: 150,000€
- General Expenses website: 50,000€
- Cannibalization: Candidate needs to calculate the number of users that will switch from paper to e-reader with Exhibits 1 & 3

PROFITABILITY CALCULATION

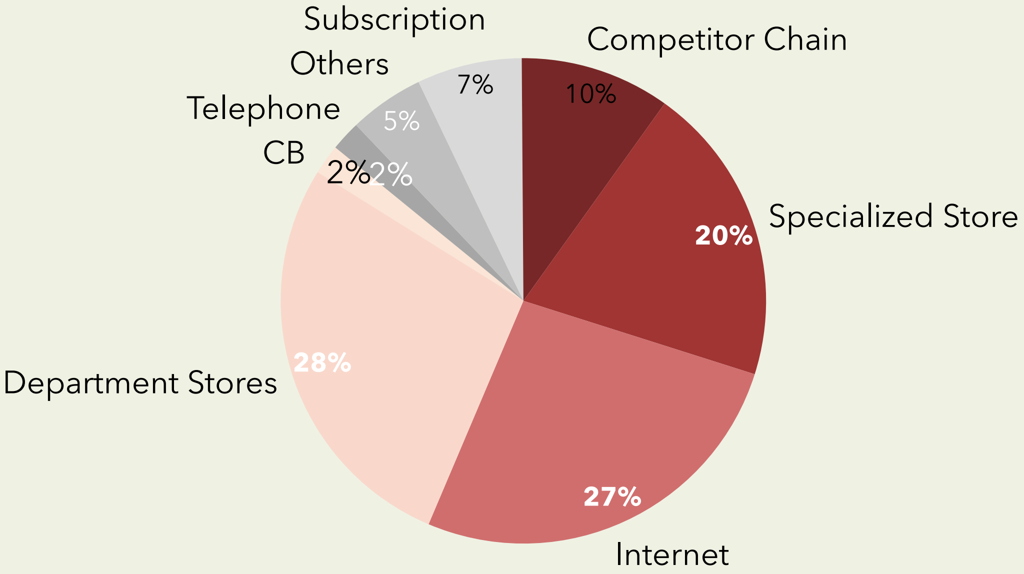
YEAR	1	2	3	4	5
Market Size	48M €	50.4M €	52.92M €	55.6M €	58.38M €
Revenues e-books	480,000 €	504,000 €	530,000 €	560,000 €	584,000 €
Gross Margin	190,000 €	201,600 €	212,000 €	224,000 €	232,000 €
Profit from e-readers	120,000 € *	3,000 € **	3,250 €	3,750 €	3,000 €
Webpage investment	- 150,000 €	-	-	-	-
General Expenses	-50,000 €	-50,000 €	-50,000 €	-50,000 €	-50,000 €
Cannibalization	-90,000 € (50,000 books x -1.8€)	-90,000 €	-90,000 €	-90,000 €	-90,000 €
TOTAL PROFIT	**20,000 €**	**64,600 €**	**75,250 €**	**87,750 €**	**95,000 €**

(*) This calculation has taken into account a gross margin of 10€ per e-reader. Candidate will have to pick the price in each case. 12,000 readers are sold the first year, 11,000 to customers switching from paper to e-book and 1,000 to new customers.

(**) No customers switch from paper to e-book after 1st year. New readers are sold to new customers acquired by market growth.

Exhibit 1: Non-Technical Books Market in Spain

PAPER BOOKS MARKET IN SPAIN

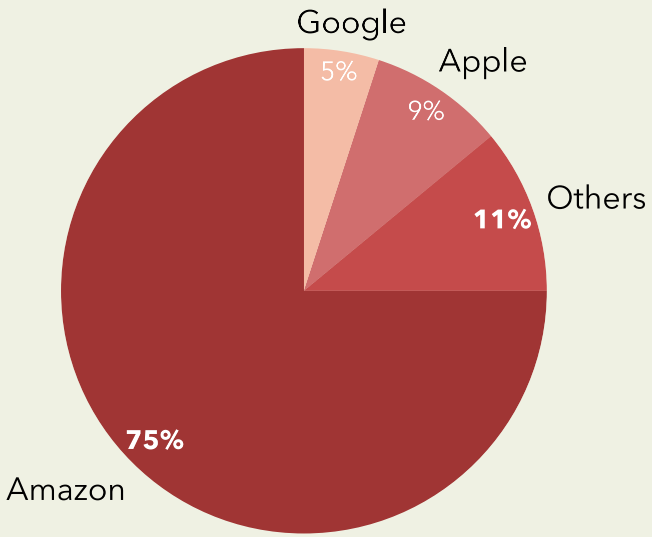


Market Size: 1,170M€

Annual Growth = 0%

Average price per book: 15€

E-BOOKS MARKET IN SPAIN



Market Size: 48M€

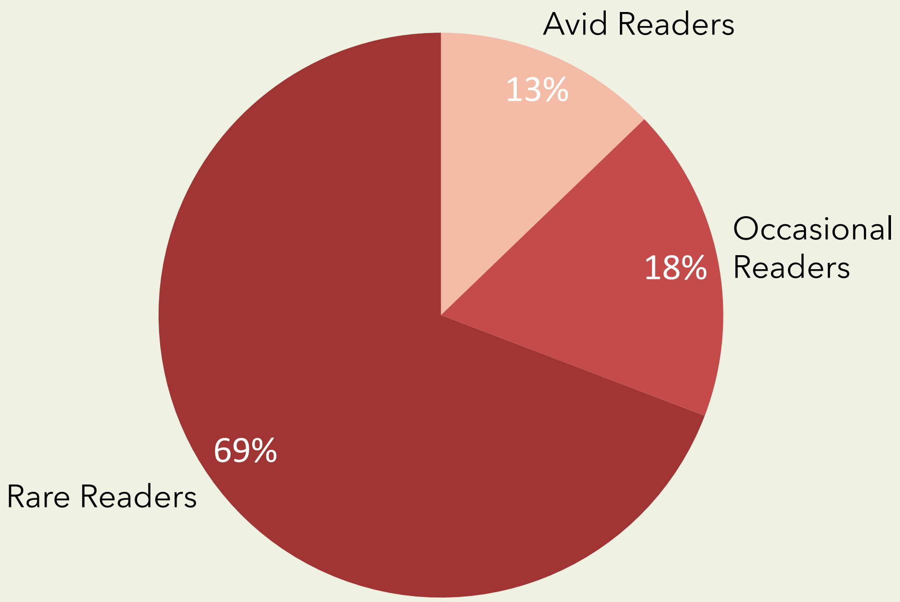
Annual Growth = 5%

Average price per e-book: 8€

Average consumption per reader: 10 e-books/year

EXHIBIT 2: CB'S CUSTOMERS INFORMATION

TOTAL CUSTOMERS: 390,000



What do you value the most about Classic Bookstore?

Would you consider switching to an electronic reader?

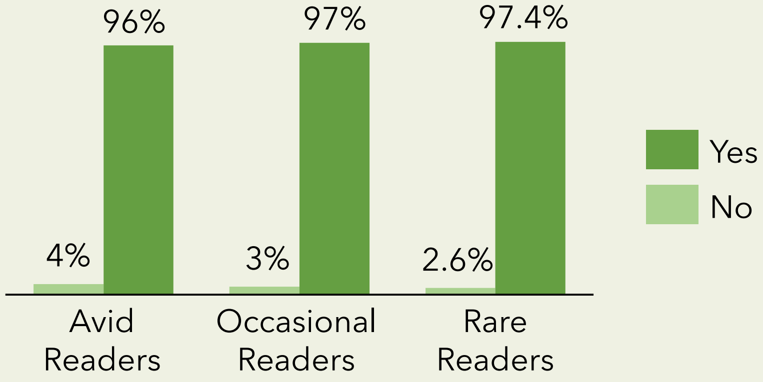


EXHIBIT 3: ELECTRONIC READERS

	CB e-Reader	Sony e-Reader	Amazon Kindle	Amazon Kindle PRO
Price	TBD	120€	100€	150€
Cost	60€	-	-	-
Formats accepted	CB	All formats	All formats	All formats
Storage	Up to 1,000 books	Up to 1,500 books	Up to 750 books	Up to 2,000 books
Extra Features	Medium	Medium	Low	High

Brainstorming 1

How would you launch this product?

Expected Brainstorming 1

In this question, the analysis should be carried out by focusing on the following main aspects:

- **Segmentation**: What users are we targeting? What could be out main target considering the company's strategy and client base?
- **Product**: How can we highlight the strong aspects of our product? What do the customers want and how can we meet their demands?
- **Price**: Although the price has been set before, additional measures can be explored, such as promotions, free gifts to great customers, etc.
- **Promotion**: How should this product be marketed? What promotions should be used?
- **Place**: Through which channels should this reader be sold?

Brainstorming 2

What other measure could our client implement in order to increase revenues?

Expected Brainstorming 2

In this question, the candidate should come up with additional measures to increase the current revenues streams. This measures should include:

- Increase number of products. Maybe including technical books in our offer could increase the number of customers.
- Creation of a loyalty scheme to try to increase average spending per customer.
- Creation a referral program to increase our customer base.
- Negotiate with e-reader supplier to include additional features.

Recommendation

What is your final recommendation for Classic Bookstore?

Sample Recommendation

- The candidate should recommend to enter in this new market of electronic readers and e-books.
- Based on our calculations and projections, our client should expect a return of 342,600€ over 5 years from an initial investment of 150,000€.
- RISKS: 1) Calculations have been based on projections of market share and a survey given by the client. Any deviations from this data could affect the profitability of the investment. 2) Similar book chains to the client could enter this market and reduce our client's potential market share. 3) Cannibalization with our client's current business model could damage the company results and image.
- POTENTIAL NEXT STEPS: These risks could be mitigated by producing a deeper market analysis and carrying out further surveys among customers to have a more accurate prediction of the market behavior. Different programs could be explored to increase customer loyalty and new customers acquisition.

Excellent candidate:

- A candidate who takes into account that taxes and time value of money have not been taken into account during the profitability analysis and would reduce the profitability of the investment.
- A candidate who mentions the lack of experience and knowledge of our client in this new business as part of the risks. This could lead to a reduction in customer service, which is very valued by our clients.

GYMCO

By Pieter Swart (IESE MBA 2021)

Sports
Wellbeing
Growth Strategy

Easy
Medium
Hard

GYMCO

Sports
Wellbeing
Growth Strategy

Easy
Medium
Hard

Prompt

Your client is an international chain of fitness centers, operating in Sub-Saharan Africa, Europe and Southeast Asia

GymCo missed its 2013 growth target of ZAR600M

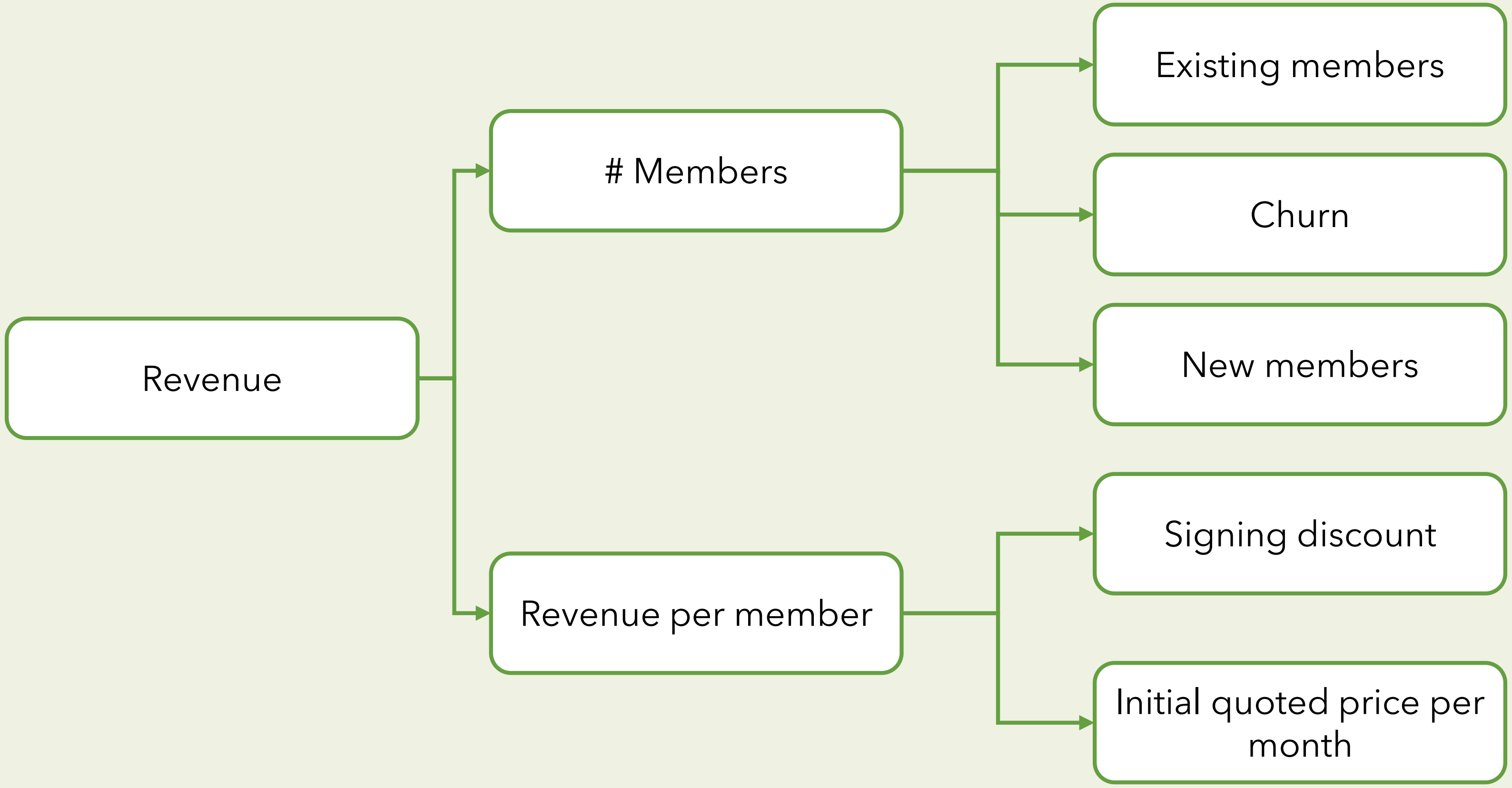
The CEO would like you to investigate what is going on

Clarifying Points (if asked)

- There are 2 major gym chains, GymCo has 60% market share, FitnessCo has 30%, and a few small chains the remaining 10%
- GymCo members pay a monthly membership fee of ~ZAR700 pm
- Market trends are in favour of gyms – consumers are switching to have more healthy habits
- There are a few smaller competitors that have recently entered the markets – these are smaller gyms offering more classes, with less focus on free weights and cardio sections
- No other competitors have noticed any decline in revenues; in fact, they have had strong increases over the past 12 months

Interviewer Guidance – Structire

Structure example



Revenue and/or Volume Analysis

Handle over Exhibit 1 if the interviewee asks for revenues and/or members volume

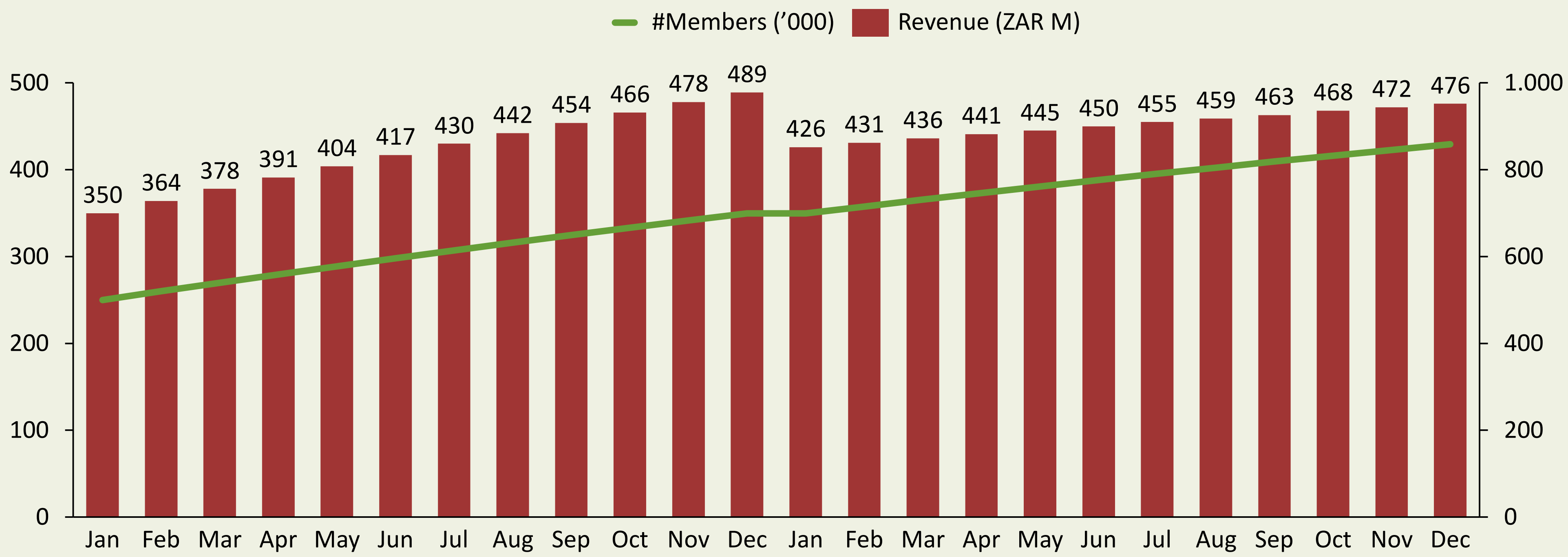
Expected Takeaways

- # members is increasing, but revenue is not increasing by the same rate– therefore revenue per member must be decreasing
- I see that churn and # of new members jump up as revenue starts declining (early 2013), did a lot of people leave and rejoin for some reason? I'd like to explore that more
- The overall takeaway is that it seems that we offered some sort of discount package that caused more people to join / rejoin

Further Info For Candidate (if asked)

- Yes, indeed – GymCo partnered with a big health insurer (HealthCo) in 2012, where they would offer 40% off to HealthCo members – the intention was to increase sign-ups (Normal price ZAR700 pm, HealthCo price ZAR400 pm)
- However, during 2013, GymCo management started noticing that some members were leaving the gym, switching healthcare providers to HealthCo, and rejoining at the new discounted rate; The estimation is that 50% of people who left the gym during 2013, rejoined at the lower rate
- Effectively, GymCo was now making less revenue of its existing members, and only some new revenue from new sign-ups

Exhibit 1: Revenue and #Members Over Time



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
# new members ('000) – end of month	30	30	30	30	30	30	30	30	30	30	30	205	30	30	30	30	30	30	30	30	30	30	30	30
Churn rate (at end of month)	2%	2%	2%	2%	2%	2%	2%	2%	2%	2%	2%	30%	2%	2%	2%	2%	2%	2%	2%	2%	2%	2%	2%	2%

Assessment of Contract Analysis

The GymCo CEO wants to know the real financial impact of the contract – has it been beneficial? If not, how much has GymCo lost as a result?

Expected Consideration

Use this to prompt candidate and frame calculation

In order to assess the decision, we need to consider the incremental revenues from the contract

Additional revenues:

- # Brand new members who joined because of the discount, and would not have joined without the discount

Foregone revenues:

- # of existing members who switched to the HealthCo discount (i.e. who would have paid R700, but now only pay R400)
- # of new members who joined on HealthCo discount who would have joined at full price irrespective of the discount

Provide Exhibit 2 when the candidate point it out the consideration above – if she/he not explain exactly this, point it out and provide Exhibit 2

Exhibit 2

2013 figures - All figures in '000	2013
# Normal members at start of year	699
# Normal members joining during year	60
# Normal members leaving during year	313
# Normal members at end of year	446
# HealthCo Discount members at start of year	0
# New HealthCo discount members during year	475
# HealthCo discount members leaving during year	66
# HealthCo Discount members at end of year	409

Calculation Info

- Normal members pay ZAR700 pm, HealthCo Discount members pay ZAR400 pm
- At the start of 2013, 200K normal members left and rejoined with HealthCo discount (i.e. they would be willing to pay the full price, but rejoined to get the R400 discount)
- Of the brand new sign-ups, 30% would have joined GymCo at full price anyway (irrespective of the discount, i.e. they would have paid ZAR700pm instead of the discounted ZAR400pm)

GYMCO

CALCULATION

How much revenue is GymCo foregoing from existing members that are switching providers?

We need to segment the # of new HealthCo members into three groups –
- **Group 1:** those that were existing members who rejoined,
- **Group 2:** Those who would have joined anyway, regardless of discount
- **Group 3:** Those that are completely new members, joining as a result of discount

Group 1: # Existing members who rejoined = 200K;
- lost revenue = 200K (700-400) pm *12 months (since they all rejoined at the start of the year) = ZAR720M revenue foregone

Group 2: # of HealthCo members who would have joined GymCo irrespective of the HealthCo deal = 475K – 200K (existing GymCo members) = 275K * 30% = 82.5K
- Revenue foregone on these members who would have joined anyway = 82.5K * (700-400) * 6 months (average duration, assuming joining throughout the year) = ZAR297M revenue foregone

Group 3: # of members acquired as a result of HealthCo deal = 475K – 200K (existing GymCo members) – 82.5M = 192.5M new members
- Additional revenue = 192.5 * R400 per month (HealthCo rate) * 6 months (average duration, assuming joining throughout the year) = ZAR462M

Therefore additional revenue of ZAR 462M minus foregone revenue of ZAR1017M (297M + 720M) = **net negative effect of 555M**
The contract is very detrimental to GymCo

Expected Insight:
Furthermore, the impact in 2013 seems to have been even greater, and this new contract with HealthCo is putting strain on GymCo's business model – too much reliance on HealthCo for new members, unable to get new members organically
Therefore, GymCo should try and renegotiate or cancel the contract

What Should GymCo do?

What are the options available to GymCo with regards to the contract?

Example Response

GymCo has a few options available:

- Get out of the deal with HealthCo
- Reduce the discount that HealthCo is giving members
- Change policies to prevent members from rejoining for a certain time period (e.g. 12 months) if they have left the gym

Option	**Benefits**	**Drawbacks**
Get out of the deal	No loss in revenue due to no more switching	• Potential loss in future # of clients • What to do about existing discounted members? If we charge full price, they might leave
Reduce discount	Discourage switching Still incentive for brand new members to join	• Likely reduced # of new members • What to do about existing discounted members? If we change their price, they might leave
Change policy	No impact on # of new sign-ups	• Long-term, perverse incentive still exists
Change to limited offer (i.e. discount only lasts for first 6 months)	Likely to get same number of new sign-ups	• Discounted clients might leave at the end of the discounted period

Recommendation

The CEO wants to meet with us in a few minutes to discuss our findings as well as the way forward – what will you tell him?

Sample Recommendation

- GymCo is currently in a very onerous contract with HealthCo, and it should cancel it. The contract was having a negative impact of ZAR555M per annum
- Furthermore, this contract places too much reliance on and gives too much power to HealthCo
- GymCo should find a way to get new members organically, while retaining existing HealthCo members
  - Offering discounts for yearly subscriptions
  - Marketing ideas
  - Offering competing classes (to compete with the smaller gyms)
  - Promotions on new members (bags / towels etc.)
- In order to decide on the best strategy, I would like to quantify the above options (cost vs. benefit)

GREEN AIRLINES

By Antonio Niemeyer (IESE MBA 2021)

Airlines
Growth Strategy
Investment Decision

Easy
Medium
Hard

GREEN AIRLINES

Airlines ···· Easy
Growth Strategy ···· Medium
Investment Decision ···· Hard

PROMPT

Due to the recent bankruptcy of a major airline, the aviation authority from Brazil recently opened an auction for landing and takeoff slots in one of the country's biggest airports. A slot is the right to land and depart from an airport during a given period of time.

Green Airlines, a small, regional airline operating in the North part of the country, was offered 10 slots, for the total price of $100M. If Green accepts to buy the slots, it would have to operate them for at least 5 years. **The owner and CEO of Green Airlines approached your firm looking for an advice on whether they should buy the slots or not.**

CLARIFYING POINTS (if asked)

- An airport slot is a permission granted by the owner of an airport designated, which allows the grantee to schedule a landing or departure at that airport during a specific time period.
- Green airlines currently does not operate in that particular airport
- Green airlines currently only does regional flights and has no plans to include international flights in its offerings
- Green airline currently does not have the necessary planes to operate the slot. Management will need to lease 5 airplanes to operate the 10 slots.
- The $100M that Green Airlines would have to pay is a one-off payment due before operations start
- The main objective of the owner/CEO is financial gain
- Green Airlines can sell the slots, but only after five years of operation
- If Green Airlines does not buy the slots, they will be sold to another airline

CASE GUIDANCE

In this case the candidate will need to lead through directive questioning. Although the answer may seem straightforward, the case will force the candidate to analyze the problem from different perspectives, and there it will demand not only math skills and problem-solving skills, but specially creativity.

A strong candidate will quickly realize that in order to answer the questioning from Green Airlines CEO, it will be necessary to understand the strategic fit and the financial implications of buying the slots. After realizing that it does not make sense to buy the slots to operate them, the candidate should explore alternative way to explore the opportunity that has emerged

GREEN AIRLINES

Interviewer Guidance – Structure

The candidate should realize that this is an opportunity for Green Airlines to expand its business into one of the country's main airport.

A suggestion of items to consider are:

Market. What's the trend for the demand of flights in the airport's region? What's the profile of travelers (business or leisure)? Are the other airlines going through financial difficulties? Is the industry suffering in general or was the bankruptcy a one-off event?

Competition. How many companies operate in that airport? Are the slots currently concentrated in the hands of a few companies or are they split among several companies? Are there Low Cost Carriers operating in the airport? What's the competition price?

Revenues/Costs. What's the expected number of passengers per day per slot (plane size, load factor, flights/day)? What's the expected price per passenger? What are the fixed costs? What are the variable costs?

Internal Capabilities. Does Green Airlines have the operational capabilities necessary to operate the slots (planes, overhead, sales system, suppliers)? Does operating in a big airport demand a different strategy than operating small, regional airports? Does Green Airlines have the financial capabilities necessary to pay for the slots? If not, what are its options?

Risks/Alternatives. Cultural issues of setting up operations in a different area. Are there other regions that may be more attractive?

GREEN AIRLINES

INTERVIEWER GUIDANCE – PART I

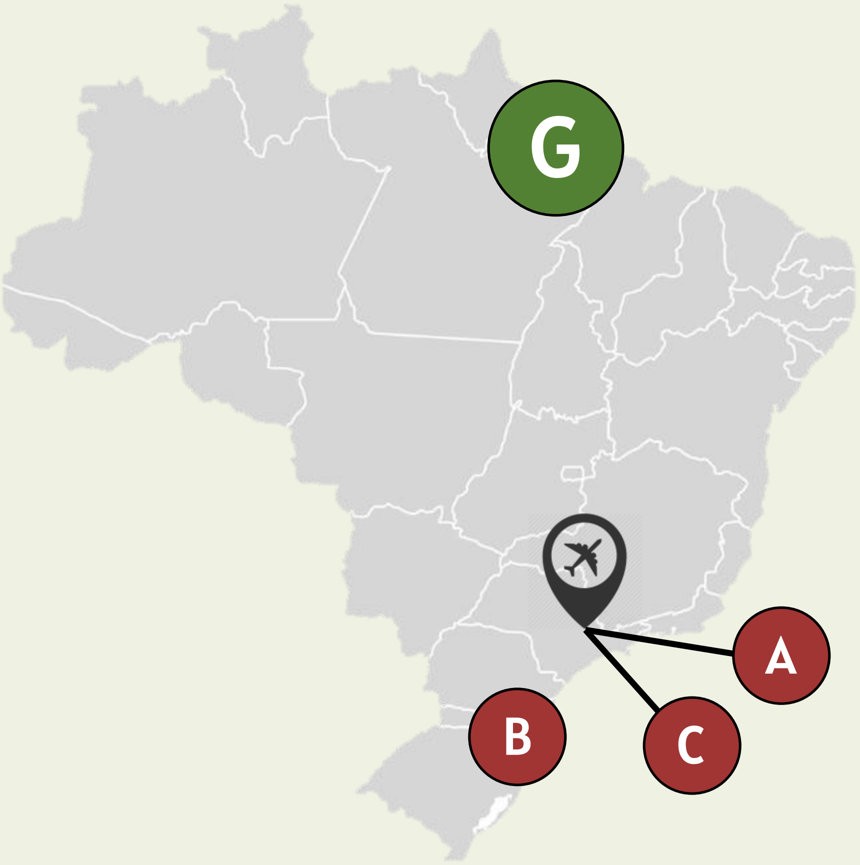
If the candidate raises concerns related to **competition, operational challenges or the aviation market in Brazil, hand Exhibit 1** (next page) to clarify these points.

KEY TAKEWAYS – EXHIBIT I

- The candidate should notice that Green Airlines operates in a very distinct region of Brazil, and is much smaller than the main players of the Sao Paulo region.
- Buying the 5 planes would mean almost doubling Green Airlines fleet, and represents an important strategic shift
- Airlines A and C are big players located in Sao Paulo, and they currently operate planes bigger than the other airlines, however, the price per ticket is lower (this may be due to shorter flights and high competition)
- The average load factor in Sao Paulo is significantly higher than the average for Green
- If Green were to buy the slots, it would most likely need bigger planes and lower prices

GREEN AIRLINES

EXHIBIT 1



Current Operations	Airline A	Airline B	Airline C	Green Airline
City Hub	São Paulo	Santa Catarina	São Paulo	Pará
Total # of planes	100	80	70	6
Avg plane size (# seats)	300	200	300	100
Avg Load Factor (%)	80%	70%	80%	60%
Avg Ticket Price ($)	200	250	200	300

Airport selling slots

A Airline A Main Hub

B Airline B Main Hub

C Airline C Main Hub

G Green Airline Main Hub

REVENUE ANALYSIS

First step is to estimate the potential revenues of the slot operation. **Discuss with the interviewee what factors he/she would use to estimate the revenues.** When asked, provide the following information in the table:

Revenues – For operation of 10 slots

# of planes	5
flights / plane / month	50 flights
Seats / plane	250 seats
Average Load Factor (%)	80%
Average Ticket Price	$ 200

REVENUE CALCULATION

Revenue per month = # of planes * # flights/plane * # seats * load factor * ticket price

= 5 * 50 * 250 * 80% * 200 = 50,000 passengers * $200/passenger

= $10 million/month

GREEN AIRLINES

COST ANALYSIS

Second, the interviewee should estimate the costs of operating the slots. **Ask him/her what he/she believes to be the main costs of an airline (fuel, crew, maintenance, insurance, leasing, fees, overhead, etc.).** After discussing the main lines of cost, provide the following information in the table:

Costs – For operation of 10 slots

Initial Investment	$5M (to set up operations)
Fuel	$14k per flight
Other Variable Costs	$4k per flight
Salaries	$1,5M per month
Maintenance & Leasing	$400k per plane/month

COST CALCULATION

Fuel: $14k * 50 * 5 = $ 3,5M/month

Other Variable Costs: $4k * 50 * 5 = $ 1,0M/month

Salaries: $ 1,5M/month

Maintenance & Leasing: $100k * 5 = $ 2,0M/month

Insurance, Fees & Others: $ 2,0M/month

Total Cost: $10M/month

Expected Profit:

Year 1: -$5M + $120M - $120M = -$5M

Following years: $120M - $120M = $0

Expected Insight:

The candidate should realize that the expected **operational result is of -$5M in Year 1, and zero in the following years** for the 10 slots on sale.

Alternative Analysis

After concluding that the operating profit would be zero for the slots, **ask the candidate which additional analyses he/she would make in order to decide to buy the slot or not.**

Potential Alternatives

The candidate should come up with potential alternatives:

Improve operational metrics.

- Possible to increase revenues (increase ticket price, include non-ticket revenues, offer packages, shuttle services, etc)?
- Possible to reduce costs (use bigger planes to reduce fixed costs, negotiate lease terms, exclude food inflight, automatization of processes, change fuel supplier, etc)?

Buy slots and sell to other company.

- Airline A and Airline C have their Hubs in Sao Paulo. The slots are probably worth a lot for them.
- How much is the market value of a Slot?

Interviewer Guidance – Part 2

After discussing the potential alternatives, **state that management has already explored all alternative ways to improve the operational result, and the numbers presented are already considering all operational improvements possible.**

Expected Takeways

If the candidate does not reach this solution alone, **say that the slots are very valuable for the big airlines operating in the region.**

The big airlines have operational advantages related to **scale**. They operate bigger planes (300 seats) than Green Airlines, so their potential revenues are higher (all other assumptions remain the same, including costs).

Ask the candidate to calculate the value of the 10 slots for the big airlines, considering planes with 300 seats.

Expected Calculation:

New Revenue per month = 5 * 50 * 300 * 80% * 200 **= $12M/month**

New Monthly profit = $2 million/month
= $24 million/year

INTERVIEWER GUIDANCE – PART 2

Now that we know the operational results for a big company utilizing these 10 slots, **ask the candidate to estimate the value of the slot for a big company such as Airline A**. Provide the following information if requested

Valuation of Slots

Discount Rate	10% per year
Right to use	perpetual

CALCULATION

Expected calculation (assuming the Net Profit as a perpetuity):

Value of slots = $24M / 10% = **$240M**

Expected Insight:

The candidate should identify that the 10 slots have a total value of approx. $ 240M for the big airlines, and that the best choice is to buy the slots, operate them for 5 years at zero profit, and sell them to a big airline for a value between $105M and $240M.

RECOMMENDATION

Ask what is the interviewer what is her/his recommendation

SAMPLE RECOMMENDATION

The recommendation should be that Green Airlines buy the 10 slots, operate them for the mandatory 5 years and then sell them at a potential profit of approx. $135M

Special Thanks

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